

Supplement S4: Sieve Diagnostics

Coincidence-Probability Tables (Archived from v3.4 §7.5) Brieuc de La Fournière
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Companion to the K_7 framework paper v3.5. This supplement archives the coincidence-probability tables from the 3.4 §7.5 “Coincidence Test” section, which the 3.5 restructure demoted from statistical headline to internal-consistency diagnostics. The archived tables remain available for readers who want the previous framing. The 3.5 headline methodology is the Sieve reading (main paper §7); see the Arithmon methodology paper (Zenodo [10.5281/zenodo.20666879](https://zenodo.org/record/20666879)) for the underlying construction.

S4.1 Purpose

The 3.4 paper reported joint coincidence probabilities

$$P_{\text{uniform}} \sim 10^{-346}, \quad P_{\text{algebraic}} \sim 10^{-133}$$

for the 33 Type I observables under (i) a uniform null model over $[0, 50\%]$ deviations, and (ii) an algebraic null model built from $4.2 \cdot 10^6$ random formulas over the same 20 structural constants. These figures do not distinguish a *genuine* survivor of a public sieve from a *well-tuned* formula grammar; the 3.5 update supersedes them by the Sieve reading of §7 as headline, and preserves them here as diagnostic evidence of internal consistency.

S4.2 Uniform Null Model (archived tables)

Multiple statistical tests under the null hypothesis of uniform deviations in $[0, 50\%]$:

Test	Statistic	df	p-value
χ^2	1063	33	5.0×10^{-202}
Fisher combined	1100.5	66	2.2×10^{-187}
KS (pull normality)	0.189	n/a	0.165
Combined uniform	–	–	10^{-346}

Pull distribution: mean = -0.774 , std = 5.62 . 72.7% of pulls within 1σ , 87.9% within 2σ . The KS test ($p = 0.165$) is consistent with Gaussian pulls: the deviations are not cherry-picked.

Reduced $\chi^2 = 32.2$ is large, driven by outliers (δ_{CP} , $\alpha_s(\text{RGE})$). The Bayes factor $\log_{10} = -2.3$ is inconclusive, reflecting this: the bulk of predictions match extremely well, but a few outliers inflate the tails. Removing the 4 known outliers gives reduced $\chi^2 < 2$.

S4.3 Algebraic Null Model (algebraic_MC, archived)

The uniform null model assumes predictions are random numbers. A stronger test: use random *algebraic formulas* from the same 20 structural constants. 4,188,086 unique formula values were generated via exhaustive depth-1/2 enumeration ($1.8 \cdot 10^6$) plus $3 \cdot 10^6$ random expression trees (depth 2–4), using 5 binary operations ($+$, $-$, $\times$, $\div$, $\wedge$) and 5 unary transforms (id, inv, $\sqrt{}$, $\square^2$, $\ln$).

Metric	K_7 framework	Random algebraic (median)	Factor
Mean deviation	0.73%	$4.1 \cdot 10^9$ %	$5.6 \cdot 10^9$
Exact matches ($< 0.01\%$)	5	0	–
Within 1%	28	0	–

Per-observable: combined $P(\text{random algebraic formula matches the framework's precision}) = 10^{-133}$ (product over 33 observables). Only 0.02% of $4.2 \cdot 10^6$ formulas match *any* observable within 0.01%.

Set-level: 0 out of 3,000,000 random sets of 33 algebraic formulas achieved the framework's mean deviation, exact count, or within-1% count. Under this null: $P < 3.3 \cdot 10^{-7}$ for each metric.

Interpretation (archived, 3.4 language). The algebraic null model is far more generous than the uniform null (it generates formulas from the *same constants*), yet the framework's performance remains unmatched across $3 \cdot 10^6$ trials. Under both declared null models, chance agreement is excluded at extreme significance: $P = 10^{-346}$ (uniform) and $P = 10^{-133}$ (algebraic). These figures quantify agreement under each declared model; they do not constitute an absolute probability that the framework is correct.

S4.4 What changed in 3.5 (Sieve reading)

The Sieve reading of §7 in the main paper isolates individual *survivors* rather than joint probabilities. Under the 4-null battery (uniform / algebraic / factorised / permuted) followed by budget-uniqueness ranking and Lean 4 R2 formal-identity flag, the outcome is:

- m_H/m_W (Route A): unique survivor at exact rank 1 budget-unique; carries the formal-identity flag via its Lean 4 R2 pre-registered identity.
- Koide's $Q = 2/3$: passes the battery with a narrow budget margin.
- n_s : passes the battery without a distinguishing budget signature.

The joint coincidence-probability language of §7.5 (this supplement) is not the right unit for the Sieve reading: two different jointness questions (probability under null vs certified survivor of a public sieve) are conflated in the 3.4 framing. The 3.5 restructure separates them and puts the certified-survivor language on the main line.

S4.5 Reproducibility

The archived null-model computations are reproducible from the same scripts used for the 3.4 release; the Sieve reading is reproducible via the Lean 4 module `Sieve/GStruct.lean` in the public `arithmon/Lean` repository. The Type I observable list is unchanged from 3.4 modulo the reconciliations documented in the CHANGELOG (observables reconciliation: 0.99% Type I after the audit against the reconciled post-freeze sources used by the Arithmon/Sieve pipeline; the Type I tables of this paper use the earlier frozen PDG 2024 + NuFIT 6.0 dataset, mean 0.73%, cf. S2 §22; m_c/m_s erratum inlined).

The K_7 Framework (formerly GIFT): Supplement S4: Sieve Diagnostics

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